

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12417521
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Jacobs; David et al.

Systems and methods for manipulation of shadows on portrait image frames

Abstract

Systems and methods described herein may relate to potential methods of training a machine learning model to be implemented on a mobile computing device configured to capture, adjust, and/or store image frames. An example method includes supplying a first image frame of a subject in a setting lit within a first lighting environment and supplying a second image frame of the subject lit within a second lighting environment. The method further includes determining a mask. Additionally, the method includes combining the first image frame and the second image frame according to the mask to generate a synthetic image and assigning a score to the synthetic image. The method also includes training a machine learning model based on the assigned score to adjust a captured image based on the synthetic image.

Inventors:	Jacobs; David (Mountain View, CA), Tsai; Yun-Ta (Mountain View, CA), Barron; Jonathan T. (Mountain View, CA), Zhang; Xuaner (Mountain View, CA)
Applicant:	Google LLC (Mountain View, CA)
Family ID:	1000008819196
Assignee:	Google LLC (Mountain View, CA)
Appl. No.:	17/786841
Filed (or PCT Filed):	December 23, 2019
PCT No.:	PCT/US2019/068266
PCT Pub. No.:	WO2021/133375
PCT Pub. Date:	July 01, 2021

Prior Publication Data

Publication Classification

Int. Cl.: G06T5/00 (20240101); G06T5/50 (20060101); G06T5/94 (20240101)

U.S. Cl.:

CPC G06T5/94 (20240101); G06T5/50 (20130101); G06T2207/20081 (20130101); G06T2207/30201 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10530991	12/2019	Wang	N/A	G06V 20/10
2012/0206470	12/2011	Frank et al.	N/A	N/A
2018/0220061	12/2017	Wang et al.	N/A	N/A
2019/0340810	12/2018	Sunkavalli	N/A	G06T 15/506
2020/0082515	12/2019	Cardei	N/A	G06F 18/214

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2020068158	12/2019	WO	G06T 5/008

OTHER PUBLICATIONS

PCT International Search Report, Application No. PCT/US2019/068266, mailed Aug. 17, 2020, 2 pages. cited by applicant

PCT Written Opinion, Application No. PCT/US2019/068266, mailed Aug. 17, 2020, 6 pages. cited by applicant

Le et al., “Shadow Removal via Shadow Image Decomposition”, IEEE, 2019, 8577-8586. cited by applicant

English translation of the First Office Action, Application No. CN2019801032852, issued Jul. 31, 2024, 13 pages. cited by applicant

Primary Examiner: Lee; Thomas D

Attorney, Agent or Firm: McDonnell Boehnen Hulbert & Berghoff LLP

Background/Summary

RELATED APPLICATION

(1) This application is the U.S. national phase under 35 U.S.C. § 371 of International Application No. PCT/US2019/068266, filed on Dec. 23, 2019, the contents of which are hereby incorporated by reference.

BACKGROUND

(2) Unless otherwise indicated herein, the materials described in this section are not prior art to the claims in this application and are not admitted to be prior art by inclusion in this section.

(3) It can be challenging to find appropriate lighting in natural environments for portrait photography. Shadows cast on faces in natural lighting environments can form corresponding photographs that are not as aesthetically pleasing as images captured with studio lighting. Conventional solutions may include relighting or otherwise manipulating the photograph using image post-processing techniques. However, it is currently difficult to automatically alter photos in a way that eliminates a shadow and still gives proper exposure to a subject's face. Moreover, in some scenarios, altering portraits can leave a high contrast shadow boundary.

SUMMARY

(4) Example embodiments relate to methods and systems for manipulating shadows on captured portrait image frames.

(5) In a first aspect, a method is provided. The method includes supplying a first image frame of a subject in a setting lit within a first lighting environment. A second image frame is also supplied. The subject of the second image frame is lit within a second lighting environment. The method also includes determining a mask and combining the first image frame and the second image frame according to the mask to generate a synthetic image. The method additionally includes assigning a score to the synthetic image and training a machine learning model based on the assigned score.

(6) In a second aspect, a method is provided. The method includes causing an image capture device to capture an image frame. The method also includes comparing the image frame to a plurality of synthetic image frames provided by a trained machine learning model. The method further includes, based on the comparison, selecting a mask from a plurality of masks associated with the trained machine learning model. The method additionally includes adjusting the image frame according to the mask to provide an adjusted image frame.

(7) In a third aspect, a system is provided. The system includes a computing device including a processor and a non-transitory computer readable medium. The non-transitory computer readable medium stores a set of program instructions provided by a trained machine learning model. The processor executes the program instructions so as to carry out operations. The operations include 1) causing an image capture device to capture an image frame; 2) comparing the image frame to a plurality of synthetic image frames provided by the trained machine learning model; 3) based on the comparison, selecting a mask from a plurality of masks associated with the trained machine learning model; 4) adjusting the image frame according to the mask to provide an adjusted image frame; and 5) displaying the adjusted image frame.

(8) Other aspects, embodiments, and implementations will become apparent to those of ordinary skill in the art by reading the following detailed description, with reference where appropriate to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. 1 illustrates a system according to an example embodiment.

- (2) FIG. 2 illustrates a system according to an example embodiment.
- (3) FIG. 3 illustrates a method according to an example embodiment.
- (4) FIG. 4A illustrates a graphical user interface according to an example embodiment in color.
- (5) FIG. 4B illustrates a graphical user interface according to an example embodiment in grayscale.
- (6) FIG. 5 illustrates a method according to an example embodiment.
- (7) FIG. 6 illustrates a data pipeline according to an example embodiment.
- (8) FIG. 7 illustrates a method according to an example embodiment.
- (9) FIG. 8 illustrates a method according to an example embodiment.
- (10) FIG. 9 illustrates a method according to an example embodiment.
- (11) FIG. 10 illustrates a method according to an example embodiment.
- (12) FIG. 11 illustrates a shadow comparison according to an example embodiment.
- (13) FIG. 12 illustrates a method according to an example embodiment.

DETAILED DESCRIPTION

(14) Example methods, devices, and systems are described herein. It should be understood that the words “example” and “exemplary” are used herein to mean “serving as an example, instance, or illustration.” Any embodiment or feature described herein as being an “example” or “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or features. Other embodiments can be utilized, and other changes can be made, without departing from the scope of the subject matter presented herein.

(15) Thus, the example embodiments described herein are not meant to be limiting. Aspects of the present disclosure, as generally described herein, and illustrated in the figures, can be arranged, substituted, combined, separated, and designed in a wide variety of different configurations, all of which are contemplated herein.

(16) Further, unless context suggests otherwise, the features illustrated in each of the figures may be used in combination with one another. Thus, the figures should be generally viewed as component aspects of one or more overall embodiments, with the understanding that not all illustrated features are necessary for each embodiment.

(17) I. Overview

(18) The present disclosure describes potential methods of training a machine learning model. For example, a machine learning model could be trained based on a plurality of image pairs, e.g., well-lit and shadowed images. This training may occur on a computing system (e.g., a cloud-based server, a desktop computer, a laptop, etc.). The trained model may then be implemented on a mobile computing device configured to capture, adjust, and/or store images. The mobile computing device may include a user interface to provide a way for users to adjust how the trained model is applied to captured images. In some embodiments, the user interface may be an application programming interface (API). In such scenarios, the interface may have graphical knobs, sliders, and/or other controls that may provide a way for a user to adjust various aspects of an input, or it may automatically adjust the input.

(19) In some embodiments, the machine learning model may be trained to “predict” alternate versions of an image with softened or removed shadows. For example, training the machine learning model may include supplying two images that share a similar or identical subject (e.g., a portrait of a human face). A first image could include the subject being lit from a bright light lighting environment. A second image could include the subject being lit with a shadow lighting environment. The shadow lighting environment could be based on a light environment that includes at least one occluder. In example embodiments, the occluder(s) could include an object or objects that may partially or completely block light that might otherwise illuminate the subject.

(20) The first well-lit image and the second shadowed image are then processed via a two-dimensional image-based data generation pipeline to create a synthetic shadowed image. The synthetic shadowed image could be used to train the machine learning model. A series of steps are taken within the data pipeline. Once a well-lit image and a shadowed image of the subject are

produced, the images can be combined in the pipeline using a mask. In an example embodiment, the mask may control the transparency/opacity of an image with respect to a background image. In some examples, a plurality of masks could correspond to different shadow shapes and/or occluder shapes. In various embodiments, the shadowed image could be layered over the well-lit image and the mask can be applied to the shadowed image. Such an arrangement could form a synthetic shadow by retaining a shadowed image portion that corresponds to a shape of the mask. In such scenarios, the remainder of the shadowed image may appear transparent so that the well-lit image is observable. In such a manner, this process forms a synthetically shadowed image. The softness and hardness of the shadow on the image can be varied by applying noise to the mask. In such a scenario, the added noise may be utilized to gradually transition between the shadowed and well-lit portions of the synthetically shadowed image.

(21) In some embodiments, the synthetically shadowed image can be assigned a score based on the quality and/or aesthetics of the image. In such scenarios, the assigned score can be used to train the machine learning model by way of reinforcement learning or other types of machine learning algorithms. This process is repeated many times (e.g., 10,000 iterations, 100,000 iterations, 1,000,000 iterations, or more) with different well-lit and shadowed images and variously shaped masks to further train the machine learning model.

(22) There are multiple ways to obtain the well-lit and shadowed image pairs. In an example embodiment, a light stage can be utilized to obtain these images. In such scenarios, the light stage, using one-light-at-a-time (OLAT) datasets, can provide arbitrary lighting environments for portrait images that share the same subject. As an example, two lighting environments may be controllably generated—one “bright” or “well-lit” lighting environment and one “shadowed” lighting environment. To provide the “shadowed” lighting environment, a region of the lighting environment or light source may be blacked out to simulate an occluder. The environments are translated into weights in the dataset which combine to produce different bright and shadow images. A second way to obtain well-lit and shadowed images is by relighting real world images (e.g., images captured in a non-OLAT, non-light stage environment). In some embodiments, an existing image is treated as the well-lit image. In such scenarios, the shadowed image could be created by 1) predicting the existing lighting environment in the all-bright image, 2) producing a “shadow” environment lighting by adding an occluder to the all-bright image, and 3) Re-rendering the image as if it had been lit by the environment in 2. Another way to obtain the all-bright and all-shadow images is by providing an image taken in a generated lighting environment and re-rendering it as described above to obtain both an all-bright and all-shadow image.

(23) In some embodiments, the shape of the mask may be the principal identifier that the machine learning model could be trained to recognize among a plurality of different shaped shadows. The shape of the mask could be used to approximate real world shapes. A shadow mask may be produced by using 3D models of occluders and facial geometry to project shadows based on features such as the contours of the face. A shadow mask may also be produced from hand drawn 2D masks, randomly synthetically generated 2D masks, or a combination of any of the above with localized color and sharpness variations to emulate real world phenomena such as subsurface scattering in skin, spatially varying lighting environments, or shadow foreshortening.

(24) The trained model may then be executed at runtime to soften, remove, or add shadows to images. For example, embodiments may include causing an image capture device to capture an image. The image may be compared to a plurality of synthetic shadowed images associated with the trained model. Based on the comparison, a shadow shape may be determined and select a corresponding shadow mask from a plurality of possible shadow masks. The captured image is then adjusted according to the shadow mask. To make a brightened image, the outline of the mask could be used to brighten the darker pixels in the shadow. To make a shadowed image, the outline of the mask could be used to lighten the pixels outside of the shadow. The shadow could also be softened by applying noise to the shadow according to the mask that matches.

(25) In some embodiments, the trained model could be stored as a set of instructions in a memory of a computing device. The trained model may be implemented in, for example, a camera app or photo editing/sharing app. The computing device may capture the image to be edited, or may provide the image from memory. In various embodiments, the trained model may also be used in a cloud-based photo management software. The user interface may utilize a knob or a slider to vary the amount of shadow or to remove one or more shadows altogether. This sort of adjustment would be automatic, semi-automatic (e.g., selectable filter presets), or fully manual. The shadow adjustment could also happen automatically without user input. In some embodiments, softening the shadows could be applied from blurring and/or applying noise to the shadow mask.

(26) By way of the described systems and methods, apparent lighting environments of images—particularly portrait images—may be adjusted to either remove or soften shadows. Such adjustments may provide a more aesthetically-pleasing image.

(27) In the present disclosure, portrait images are primarily described. Portrait images can be easily lit with a light stage, and the learning model may have an easier time during the training phase because human portrait subjects have similar facial geometry. However, it will be understood that the systems and methods described herein could be performed on other types of subjects (e.g., other types of objects, buildings, etc.) to remove and/or otherwise adjust apparent shadows. The approaches disclosed could additionally or alternatively be used to vary coloring and contrast in an image.

(28) II. Example Systems and Methods

(29) FIG. 1 illustrates a system **100**, according to an example embodiment, which implements a trained machine learning model in order to manipulate shadows on portrait photographs. The system **100** includes a computing device **102**. As illustrated in FIG. 2, computing device **102** could include any of a mobile computing device **204** (e.g. a smartphone or another type of digital camera), a laptop **206**, a cloud-based computing device **208**, or a desktop computer **210**.

(30) In some embodiments, the computing device **102** could include a processor **110** operable to execute program instructions **114**. The processor **110** may include, for instance, an application-specific integrated circuit (ASIC) or a field-programmable gate array (FPGA). Other types of processors, computers, or devices configured to carry out software instructions are contemplated herein.

(31) The program instructions **114**, executed by the processor **110**, are stored in a non-transitory computer readable medium **112**. At least some of the program instructions **114** could be developed and provided by a trained machine learning model, as described herein. The non-transitory computer readable medium **112** may be part of a memory and may be, but is not limited to, read-only memory (ROM), programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), non-volatile random-access memory (e.g., flash memory), a solid state drive (SSD), a hard disk drive (HDD), a Compact Disc (CD), a Digital Video Disk (DVD), a digital tape, read/write (R/W) CDs, R/W DVDs, etc.

(32) The image capture device **108** could be selected from a wide variety of different devices. For example, the image capture device **108** may include, but is not limited to, a smartphone, a mobile device, a wearable device, a laptop, a camera phone, or a digital single lens reflex (DSLR) camera. As an example, the computing device **108** should have the ability to take a photo, display photos, and/or access photos from computer readable medium or a cloud-based computing device. It will be understood that other types of image capture devices **108** are possible and contemplated herein to capture the image frame **1106**.

(33) The program instructions **114** may be executed by the processor **110** in the computing device **102** so as to carry out certain operations. FIG. 3 illustrates a flow diagram **300**, according to an example embodiment of the operations. The flow diagram **300** may include various blocks or steps that represent the operations. The blocks or steps may be carried out individually or in

combination. The blocks or steps may be carried out in any order and/or in series or in parallel.

(34) Block **302** includes causing an image capture device (e.g., image capture device **108**) to capture an image frame (e.g., image frame **1106**). FIG. **10** illustrates the different manners in which the image frame (e.g., image frame **1106**) may be captured. For example, capturing an image frame **1106** by way of the image capture device (e.g., image capture device **108**) may be accomplished in a plurality of ways including, but not limited to, capturing the image frame with a camera **1002**, opening an image frame file **1004**, or accessing the image frame by way of a cloud-based computing device **1006**. In some embodiments, the image capture device **108** may be part of the computing device **102**. As an example, if the computing device **102** is a mobile computing device **204**, the image frame **1106** to be manipulated could be captured by taking a photo with a camera in the mobile computing device **204**, the image frame **1106** to be manipulated could be accessed from computer readable medium **112**, or the image frame **1106** could be accessed from a cloud-based computing device **208**. It will be understood that other types of computing devices **102** are possible and contemplated herein to capture the image frame **1106**.

(35) Block **304** includes comparing the image frame **1106** that was captured above to a plurality of synthetic images (e.g., synthetic images **608**) provided by the trained machine learning model. The plurality of synthetic images **608** are generated by combining shadowed images **604** and well-lit images **602** using a mask **606** to create a partially shadowed synthetic image **608**.

(36) Block **306** then includes, based on the comparison, selecting a mask shape (e.g., mask shape **802**) from a plurality of masks **605** associated with the trained machine learning model. The mask **606** is selected from the different masks **606** used to create the partially shadowed synthetic image **608**. As illustrated in FIG. **11**, the shape of the selected mask **1104** is one that most closely fits the shape of at least a portion of a shadow **1102** within an image frame **1106**.

(37) Block **308** includes adjusting the image frame **1106** according to the mask **606** to provide an adjusted image frame **1202**. FIG. **12** is an illustration of an example embodiment of how the adjustment may occur. The mask **606** is used when adjusting to be able to determine the shape and size of the shadowed area and adjust a brightness of the shadowed area, and or adjust the brightness of the well-lit area outside of the mask **606**. The shadow **1102** in the image frame **1106** may be sharpened (e.g., enhanced), softened (e.g., reduced), or eliminated completely.

(38) Block **310** includes that once the image frame **1106** is adjusted, the adjusted image frame **1202** is displayed. For example, the adjusted image frame **1202** could be displayed on the display **106** of the computing device **102**. Additionally or alternatively, the adjusted image frame **1202** may be displayed using a graphical user interface **104**.

(39) There are different ways to achieve an adjusted image frame **1202**. An adjusted image frame may be obtained by using the above steps in a controllably adjusted manner. For example, the user may make the adjustments to the photo by manually selecting the options in a graphical user interface **104** so the user controls how much of an adjustment is made. Alternatively, the adjusted image frame **1202** may be automatically adjusted by the computing device **102**. For example, a user may not have any input on the amount of adjustment. The computing device **102** may make the determination of how much and what kind of adjustment to make.

(40) Further, the graphical user interface **104** may include a control interface **402**. FIGS. **4A** and **4B** illustrates the graphical user interface **104** according to an example embodiment. While FIGS. **4A** and **4B** illustrate a graphical user interface **104** on a mobile computing device **204**, it will be understood that other types of computing devices **102** and other types of graphical user interfaces are possible and contemplated herein. The control interface **402** may display the image frame **1106** in the center with user input options surrounding the image frame **1106** to allow the image to be re-lit. As an example, the user input options could appear as user-operable effect icons along a bottom portion of the control interface **402**. The control interface **402** may be configured to controllably adjust the image frame **1106** according to the selected mask **606** and/or the user-operable effect icons. The image frame **1106** is controllably adjusted by changing the appearance of the shadow

1102 on the image frame **1106**. Along with other possible and contemplated color changes, the shadow **1102** could be controllably adjusted by sharpening the shadow **1102**, softening the shadow **1102**, or removing the shadow **1102** altogether.

(41) In the graphical user interface **104**, controllably adjusting the image frame **1106** may be accomplished by tuning knobs **404**. The tuning knobs **404** may be controllably operable to soften, sharpen, or eliminate shadows **1102** within the image frame **1106**. In some embodiments, the direction the tuning knob **404** is turned and the amount the tuning knob **404** is turned could determine how much the image frame **1106** is adjusted.

(42) Alternatively, adjusting the image frame **1106** in accordance with the mask **606** could be executed automatically by the computing device **102**. The adjusted image frame **1202** would be displayed without controlled adjustment or a control interface **402**. In another embodiment, the image frame **1106** could be automatically adjusted and then additionally controllably adjusted through a control interface **402** of a graphical user interface **104**.

(43) FIG. 5 illustrates a method **500**, according to an example embodiment. The method **500** may include various blocks or steps. The blocks or steps may be carried out individually or in combination. The blocks or steps may be carried out in any order and/or in series or in parallel. Further, blocks or steps may be omitted from or added to method **500**.

(44) While examples herein may describe shadow manipulation on image frames **1106** with subjects **614** that are portraits (e.g., individual human subjects), it will be understood that the described systems and methods could be applied to other subjects. For example, image frame **1106** could include other types of photographic subjects.

(45) Block **502** includes supplying a first image frame **602** of a subject **614** in a setting **616** lit within a first lighting environment. The first image frame **602** may be obtained in a plurality of manners. This is illustrated in FIG. 7. In an example embodiment, the first image frame **602** may be obtained from a light stage **702**. In such scenarios, the light stage **702**, using one-light-at-a-time (OLAT) datasets, can provide arbitrary lighting environments for portrait images that share the same subject **614**. As an example, different lighting environments may be controllably generated. The environments are translated into weights in the dataset which combine to produce different bright and shadow images. Alternatively, the first image frame **602** could be a real world image. A real world image is an image captured in a non-OLAT, non-light stage environment. Further still, the first image frame **602** may be an image taken in a generated lighting environment and re-rendering it.

(46) The first image frame **602** is lit within a first lighting environment. The first lighting environment is a well-lit lighting environment **610**. A well-lit lighting environment **610** is a lighting environment in which the subject **614** is lit as bright as or brighter than the setting **616**. The setting **616** is the subject's **614** surroundings. The well-lit lighting environment **610** does not contain saturated pixels and makes the features of the subject **614** clearly visible and discernable.

(47) Put another way, a well-lit lighting environment **610** could include considerations based on the camera system (e.g., aperture size, exposure time, etc.) and an image capture scenario or setting. For instance, in an example setting, capturing a “well-lit” image of a subject's face while the subject is standing in front of a sunset could require a fill-flash or another type of additional lighting. In another setting, capturing a “well-lit” image of a subject's face during the day while the subject is indoors and next to a window might not require additional lighting to produce a “well-lit” image. Accordingly, it will be understood that a well-lit lighting environment **610** could be a lighting environment in which the subject is lit sufficiently with respect to the remainder of the scene such that the subject is readily discernible, and clear, and well within the useful dynamic range of the camera.

(48) The well-lit lighting environment **610** can be obtained with a light stage **702**, as the light stage **702** can controllably generate a “bright” or “well-lit” lighting environment **610**. The well-lit lighting environment **610** can also occur in a real world scenario. Alternatively, the well-lit

environment **610** can be generated. The well-lit environment **610** might be generated in the light stage, or with professional photography lighting so that a number of lights is turned on to satisfy that the subject **614** is as bright or brighter than the subject's surroundings. Alternatively, the well-lit environment **610** may be generated by re-lighting a previously taken image. Further still, in other embodiments the well-lit environment **610** might be an environment lit by the sun with no blocking objects at a high light time. For example, 12:00 pm.

(49) The subject **614** in the first image frame **602** may be a person in a portrait image. Portrait images can be easily lit with a light stage **702** and a machine learning model may have an easier time during the training phase because human portrait subjects **614** have similar facial geometry **812**. However, it will be understood that the systems and methods described herein could be performed on other types of subjects, such as buildings, other types of objects, or other living things, to remove and/or adjust apparent shadows.

(50) Once the first image frame **602** is obtained, it is then supplied by adding it to a data pipeline **600**. The first image **602** is added to the data pipeline **600** to combine with the second image frame **604** and mask **606** in order to create a synthetic image **608**.

(51) Block **504** includes supplying a second image frame **604** of the subject **614** within a second lighting environment. The second image **604** is supplied by adding it to the data pipeline **600**. The second image frame **604** may be obtained in a plurality of manners as illustrated in FIG. 7. In an example embodiment, the second image frame **604** may be obtained from a light stage **702**. As previously explained, the light stage **702**, using OLAT datasets, can provide arbitrary lighting environments for portrait images that share the same subject **614**. The light stage **702** can be used to controllably generate different lighting environments. The environments are translated into weights in the datasets which combine to produce different bright and shadow images.

Alternatively, the second image frame **604** could be obtained by relighting a real world image. Further still, the second image frame **604** may be obtained by re-rendering an image taken in a generated lighting environment

(52) The second image frame **604** is lit within a second lighting environment. In an example embodiment, the second lighting environment could be a shadowed lighting environment **612**. The shadowed lighting environment **612** can be obtained with a light stage **702** by simulating an occluder. An occluder is an object that partially or completely impedes the amount of light reaching the eye or the image capture device. To simulate an occluder, a region of the lighting environment or light source may be blacked out. The shadowed lighting environment **612** can also be obtained by relighting real world images. In this scenario, an existing image is treated as the well-lit image and the shadow image could be created by 1) predicting the existing lighting environment in the all-bright image, 2) producing a “shadow” environment lighting by adding an occluder to the all-bright image, and 3) Re-rendering the image as if it had been lit by the environment in 2).

Alternatively, the shadow lighting environment **612** can be obtained by re-rendering an image taken in a generated lighting environment as described in the steps above.

(53) The subject **614** in the second image frame **604** could be a variety of different kinds of subjects such as a portrait of a person, buildings, other types of objects, or other living things, as is the case for the first image frame **602**. However, the subject **614** in the second image frame **604** must be the same or similar subject **614** as the subject **614** in the first image frame **602** in the same or similar orientation.

(54) Once the second image frame **604** is obtained, it is then supplied by adding it to the data pipeline **600**.

(55) Block **506** includes determining a mask **606**. The mask **606** creates the synthetic shadow **1102** and is created by selecting a mask shape **802**. As illustrated in FIG. 8, the mask shape **802** is selected from a plurality of possible shadow shapes **804**. The shadow shapes could be gathered from projected three-dimensional models of occluding objects **806**. The projected three-dimensional models of occluding objects **806** are projected according to a facial geometry **812** of

the subject **614**. For example, the projection could be obtained by placing an object, such as a branch, hat, or other object (or object type) which typically causes shadows, in front of a light source so that it blocks an amount of light from reaching the subject's **614** face. These shadow shapes would be projected in a way that respects the contours of the subject's **614** face. In another example, the shadow shapes gathered from projected three-dimensional models of occluding objects **806** could also be taken from the shadows on a face caused by the subject's **614** own features such as a nose.

(56) In some embodiments, the shadow shapes could additionally or alternatively be gathered from two-dimensional shapes **808**. For example, the two-dimensional shapes **808** could be hand drawn shapes of shadows, they could be random shapes, they could be traced shapes, they could include outlines from photographs, or they could be produced from any other manner not mentioned in order to obtain a two dimensional shape of a shadow. Additionally or alternatively, the shadow shapes could be gathered from synthetically-generated two-dimensional shapes **810**. The shadows could be synthetically-generated by a computer program or any other manner. Furthermore, the two dimensional shadow shapes could be a combination of any of the above possible methods with localized color and sharpness variations to emulate real world phenomena such as subsurface scattering in skin, spatially varying lighting environments, or shadow foreshortening.

(57) Block **508** includes combining the first image frame **602** and the second image frame **604** according to the mask **606** to generate a synthetic image **608**, as illustrated by FIG. **6**. In an example embodiment, the mask **606** may control the transparency or opacity of the second image frame **604** with respect to the first image frame **602**. As previously mentioned, the mask **606** corresponds to different shadow shapes.

(58) The first image frame **602** and second image frame **604** are combined to produce the synthetic image **608** by layering the second image frame **604** over the first image frame **602** and applying the mask **606** to the shadowed image frame **604**. This combination forms a synthetically shadowed image by retaining a shadowed image portion that corresponds to the shape of the mask **606** and making the remainder of the second image frame **604** appear transparent so that the first image frame **602** is observable. Some or all of the operations of combining the first image frame **602** and second image frame **604** according to the mask **606** to generate a synthetic image **608** could be performed in at least a portion of a data pipeline **600**.

(59) Block **510** includes assigning a score to the synthetic image **608**. The score assigned may be associated with the realistic look of the shadow **1102** on the synthetic image **608**. Score values could be based on a scoring range. As an example, the scoring range may start at zero and end at ten. A synthetic image **608** with a realistic looking shadow **1102** would receive a high score on the zero to ten scoring range, for example higher than five, while a synthetic image **608** with an unrealistic looking shadow **1102** may receive a low score on the zero to ten scale, for example lower than 5. The scale used for scoring the synthetic images **608** could be any range of other numbers, for example zero to 100, or zero to 200. Synthetic images **608** may be scored objectively by comparing them to a “ground truth” image that defines what a perfect scoring image would look like. Synthetic images **608** are scored by computing the “distance” between the synthetic image **608** and the ground truth image. Distance may be calculated by subtracting pixel values of one image from another, or by a variety of other techniques for distance calculation.

(60) Block **512** includes training a machine learning model based on the assigned score of the synthetic image **608**. Assigning the score to the synthetic image **608** assists with training the machine learning model. Training the machine learning model based on the assigned scores of the respective synthetic images provides a way to tune variables such as parameter weights in a convolutional neural network. Other score-based machine learning techniques and algorithms are contemplated and possible. Once trained, the machine learning model will be able to identify which synthetic images **608** are the best to reference. The machine learning model can thus create more accurate synthetic images **608** based off of prior instances. Using scoring to train the machine

learning model also improves the aesthetic qualities of adjusted image frames having softened or removed shadows.

(61) FIG. 9 illustrates a method **900**, according to an example embodiment. The method **900** may include various blocks or steps. The blocks or steps may be carried out individually or in combination. The blocks or steps may be carried out in any order and/or in series or parallel. Further, blocks or steps may be omitted from or added to method **900**.

(62) Some or all of the blocks of method **900** may involve the elements of system **100** and/or the graphical user interface **104** as illustrated and described in reference to FIGS. 4A and 4B.

Furthermore, some or all blocks of method **900** may be carried out by the computing device **102**.

(63) While examples herein may describe shadow manipulation on image frames with subjects **614** that are portraits, it will be understood that the described systems and methods could be applied to other photographic subjects.

(64) Block **902** includes causing an image capture device **108** to capture an image frame **1106**. The image capture device **108** capturing an image frame **1106** may be accomplished in a plurality of ways including, but not limited to, capturing the image frame **1106** with a camera **1002**, opening an image frame file **1004**, or accessing the image frame by way of a cloud-based computing device **1006**. The image capture device **108** may be part of the system **100** and more specifically the computing device **102**. As an example, if the computing device **102** is a mobile computing device **204**, the image frame **1106** to be adjusted could be captured by taking a photo with a camera in the mobile computing device **204**, the image frame **1106** to be adjusted could be accessed in memory, or the image frame **1106** could be accessed from a cloud-based computing device **208**. It will be understood that other types of computing devices **102** are possible and contemplated herein to capture the image frame **1106**. The captured image is the image that the machine learning model will be applied to in order to eliminate or adjust shadows.

(65) Block **904** includes comparing the image frame **1106** to a plurality of synthetic image **608** provided by a trained machine learning model. In some embodiments, the image frame **1106** could be captured in a manner previously described. As described in method **500**, the plurality of synthetic images **608** are generated by combining shadowed images **604** and well-lit images **602** of the same subject **614** using a mask **606**. Combining the shadowed images **604** and well-lit images **602** creates a partially shadowed synthetic image **608**. The two images are then compared by analyzing the shadows on the captured image frame **1106** against the shadows on the synthetic image **608**. The synthetic image **608** with shadows that most closely correspond to the shadows on the captured image frame **1106** could be selected. In some embodiments, the trained machine learning model could be similar or identical to that of method **500**, but it is also understood that other methods for machine learning models are possible and contemplated herein.

(66) Block **906** includes, based on the comparison, selecting a mask **606** from a plurality of masks **605** associated with the trained machine learning model. The mask **606** is selected from the plurality of masks **605** used to create the partially shadowed synthetic images **608**. The selected mask **606** is the mask **606** that was used to create the matching synthetic image **608**. Therefore, the shape of the selected mask **1104** substantially matches a shape of at least a portion of a shadow **1102** within the image frame **1106**, as illustrated in FIG. 11.

(67) Block **908** includes adjusting the image frame **1106** according to the mask **606** to provide an adjusted image frame **1202** with less of a shadow and an adjusted image frame **1212** with more a shadow. FIG. 12 is an illustration of an example embodiment of how the adjustment may occur. The mask **606** is used when adjusting to determine the shape and size of the shadowed area and adjust brightness of the shadowed area, and/or adjust the brightness of the well-lit area outside of the mask **606**. Adjustments to the image are made based on an image adjustment value **1204** to soften the shadow and image adjustment value **1214** to sharpen the shadow. The image adjustment value **1204** is selected so as to soften, or remove the shadow **1102** within the image frame **1106**. The image adjustment value **1214** is selected so as to sharpen the shadow **1102** within the image

frame **1106**. The image adjustment value **1204** may be a noise value such as Perlin noise where increasing the noise value **1208** will soften the shadow **1102** within the image frame **1106**. Alternatively, the image adjustment value **1214** may be a noise value where decreasing the noise value **1206** will sharpen the shadow **1102** within the image frame **1106**. In some embodiments, the noise value **1208** could be increased enough, then the shadow **1102** may be so softened that it is removed.

(68) Because of these adjustments, the adjusted image frame **1202** has less of a shadow **1102** than the image frame **1106** where the adjusted image frame **1212** has more of a shadow **1102** than the image frame **1106**. In addition to shadows **1102**, the mask **606** may also be used to change the color, saturation, or other effects of the image frame **1106**. The mask **606** could be used in conjunction with different photo editing systems to vary the effects of the adjusted image frame **1202** or adjusted image frame **1212**. Effects of the image frame **1106** that can be changed may include frequency, contrast, opacity, color balance, hue, etc.

Claims

1. A method, comprising: causing an image capture device to capture an image frame; comparing a shadow shape in the image frame to a plurality of synthetically shadowed image frames provided by a trained machine learning model; based on the comparison, selecting a mask shape from a plurality of masks associated with the trained machine learning model, wherein the shape of the selected mask is one that most closely fits the shadow shape in the image frame; and adjusting the image frame according to the mask shape to provide an adjusted image frame.
2. The method of claim 1, wherein causing the image capture device to capture an image frame comprises capturing the image frame with a camera, opening an image frame file, or accessing the image frame by way of a cloud-based computing device.
3. The method of claim 1, wherein the mask shape substantially matches a shape of at least a portion of a shadow within the image frame.
4. The method of claim 1, wherein the adjusted image frame has more or less of a shadow than the image frame.
5. The method of claim 4, wherein an image adjustment value is selected so as to soften, sharpen, or remove the shadow within the image frame.
6. The method of claim 5, wherein image adjustment value comprises a noise value, wherein increasing the noise value will soften the shadow within the image frame and decreasing the noise value will sharpen the shadow within the image frame.
7. A system, comprising: a computing device including a processor and a non-transitory computer readable medium wherein the non-transitory computer readable medium stores a set of program instructions provided by a trained machine learning model, wherein the processor executes the program instructions so as to carry out operations, the operations comprising: causing an image capture device to capture an image frame; comparing a shadow shape in the image frame to a plurality of synthetically shadowed image frames provided by the trained machine learning model; based on the comparison, selecting a mask shape from a plurality of masks associated with the trained machine learning model, wherein the shape of the selected mask is one that most closely fits the shadow shape in the image frame; adjusting the image frame according to the mask shape to provide an adjusted image frame; and displaying the adjusted image frame.
8. The system of claim 7, wherein the computing device comprises at least one of: a mobile computing device, a laptop, a cloud-based computing device, or a desktop computer.
9. The system of claim 7, wherein causing the image capture device to capture an image frame comprises capturing the image frame with a camera, opening an image frame file, or accessing the image frame by way of a cloud-based computing device.
10. The system of claim 7, further comprising a graphical user interface, wherein the graphical user

interface comprises a control interface configured to controllably adjust the image frame according to the selected mask.

11. The system of claim 10, wherein the control interface comprises tuning knobs that are controllably operable to soften, sharpen, or eliminate shadows within the image frame.

12. The system of claim 7, wherein adjusting the image frame according to the mask shape is done automatically.
